

Data Modeling: Classification

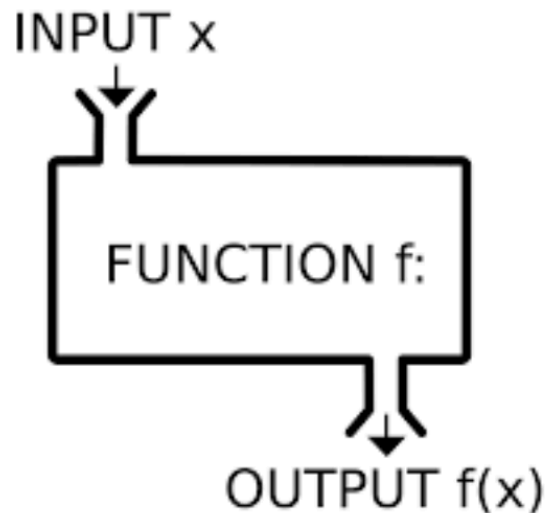
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Develop a Classification "Model"

Let's Build a Classification "Model"

We've explored the data. Now, let's use our insights to make predictions!



Our Goal:

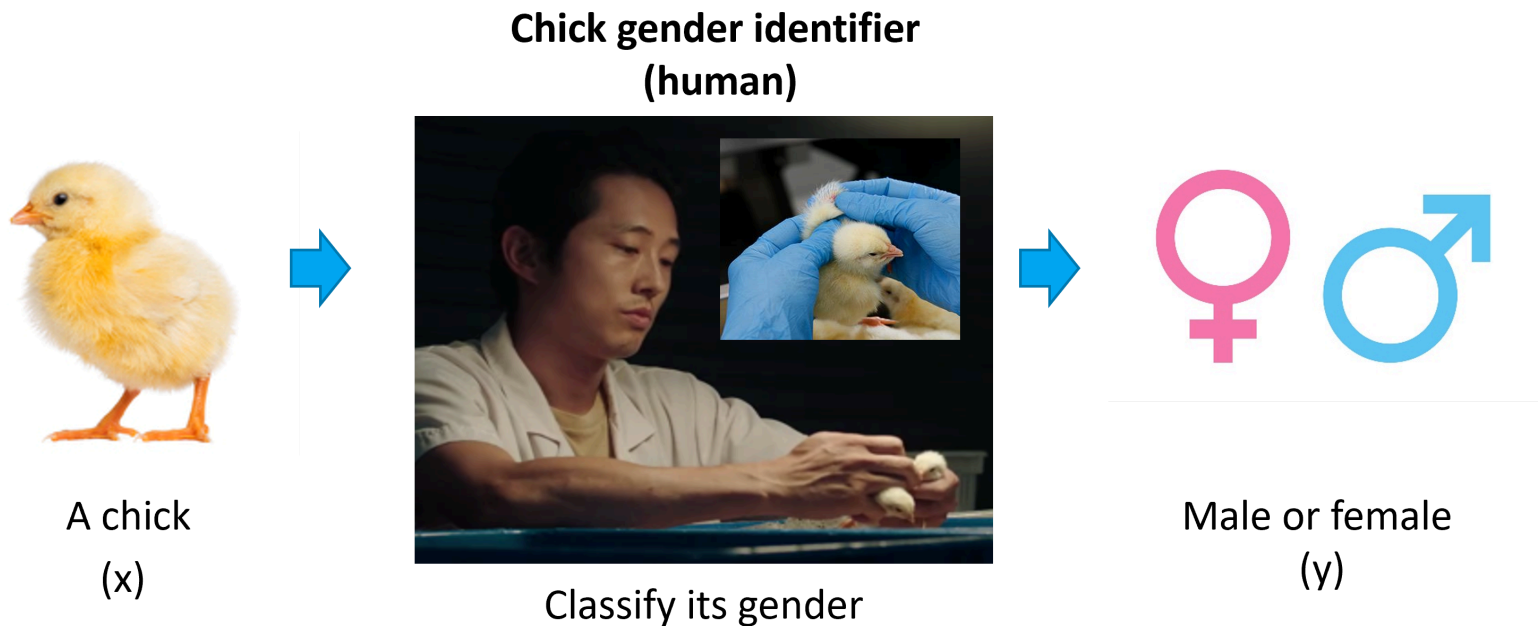
Can we build a function (f) to classify the Species (y) of an iris, just by looking at its measurements (x)?

This process is called **Classification**.

Example: Chick gender classification (by Human)

Classification is the process of categorizing items into predefined categories or **classes**.

- Input (x): several features of a chick (feather length, wing size, ...)
- Output (y): chick's gender



Example: Chick gender classification (by Machine)

Classification is the process of categorizing items into predefined categories or **classes**.

- Input (x): several features of a chick (feather length, wing size, ...)
- Output (y): chick's gender



We can make our computer to classify Iris flower into one of the predefined classes!

Components of Classification

- Input (`x` , Features): The measurements we have (e.g., `PetalWidthCm`).
- Model (`f` , Our Rule): A function we will write.
- Output (`y` , Class): The "Category" we predict (e.g., `setosa` , `versicolor` , `virginica`).

We will use `.apply()` to build our model!

"Ground Truth" vs. "Prediction"

To see if our model is any good, we need to compare its guesses to the *correct* answers.

- **Ground Truth (or True Label):**

- The *correct* answer. The "answer key."
- In our dataset, the `Species` column is our **Ground Truth**. A human expert has already labeled these for us.

- **Prediction:**

- The model's *guess*.
- We will create a **new column** (e.g., `MyPrediction`) to store our model's guesses.

Our goal is to make our `MyPrediction` column match the `Species` column as closely as possible.

Our "Models" Will Be Functions

A "model" sounds complicated, but for us, it will just be a Python function that contains our rules.

We will write two types of models:

1. Univariate Model:

- Uses only **one feature** to make a decision.
- *Example Rule:* IF PetalLengthCm < 2.5 THEN predict 'Iris-setosa'

2. Multivariate Model:

- Uses **two or more features** to make a decision.
- *Example Rule:* IF PetalLengthCm < 4.8 AND PetalWidthCm < 1.5 THEN predict 'Iris-versicolor'

Goodness of the Model: Accuracy

The most common way to measure performance is **Accuracy**.

Formula:

$$\text{Accuracy} = (\text{Number of Correct Guesses}) / (\text{Total Number of Guesses})$$

- Accuracy is just like a test score.
- It measures what *fraction* of the guesses our model got right.

Example:

- Our dataset has 150 rows.
- Our model makes its predictions.
- We compare our predictions to the "Ground Truth" (`Species`) column and find we got **141** correct.
- **Accuracy = 141 / 150 = 0.94 (or 94%)**

Thank You.